

AT2-C02

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1. Ride-Sharing Company using Reinforcement Learning

RL Framework

State Space (S):

- Driver location
- Passenger location
- Traffic condition
- Time of day
- Driver availability

Action Space (A):

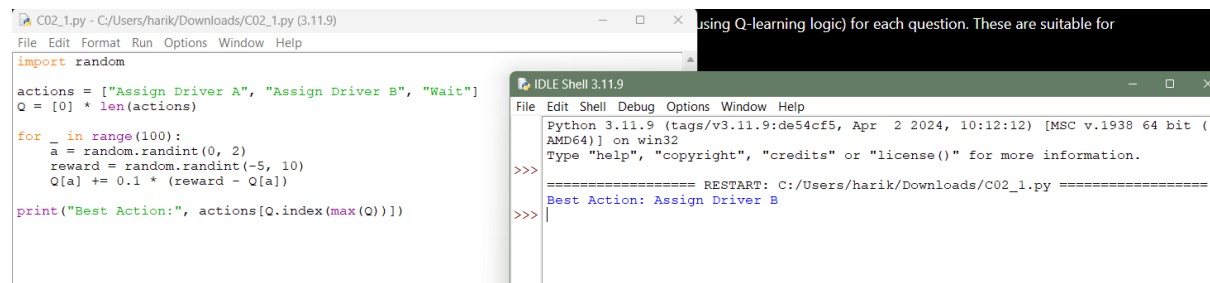
- Assign Driver A
- Assign Driver B
- Wait for nearby driver
- Reject request

Reward Function (R):

- +10 for quick pickup
- +15 for successful trip
- -10 for long waiting time
- -15 for cancelling ride

Explanation

The RL agent learns which driver should be assigned to minimize passenger waiting time and maximize driver utilization. Over time, it improves service quality and customer satisfaction.



The image shows a screenshot of a Python script in a text editor and its execution in an IDLE Shell. The script defines a list of actions, initializes a Q-table, and simulates a random environment. The shell output shows the script running and printing the best action.

```
import random
actions = ["Assign Driver A", "Assign Driver B", "Wait"]
Q = [0] * len(actions)

for _ in range(100):
    a = random.randint(0, 2)
    reward = random.randint(-5, 10)
    Q[a] += 0.1 * (reward - Q[a])

print("Best Action:", actions[Q.index(max(Q))])
```

Best Action: Assign Driver B

2. E-Learning Platform (Exploration vs Exploitation)

Exploration

- Recommends new study materials.
- Helps discover better learning resources.
- Prevents repetitive recommendations.

Exploitation

- Recommends materials the student already performs well with.
- Improves immediate learning outcomes.
- Uses previous learning history.

Balance

A balance between exploration and exploitation helps students receive both familiar and new content, improving overall learning effectiveness.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
File Edit Format Run Options Window Help
import random
materials = ["Math", "Science", "AI"]
epsilon = 0.2
if random.random() < epsilon:
    print("Explore:", random.choice(materials))
else:
    print("Exploit:", materials[0])

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
```

3. Robotic Vacuum Cleaner

RL Framework

State

- Current position
- Battery level
- Dirt detected
- Obstacle nearby

Action

- Move forward
- Turn left
- Turn right
- Clean
- Return to charging station

Policy

A policy determines the best action for every situation.

Value Function

The value function estimates future rewards from each state. As learning continues, the robot chooses cleaner and shorter paths while avoiding obstacles efficiently.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
File Edit Format Run Options Window Help
actions = ["Forward", "Left", "Right", "Clean"]
value = [5, 2, 3, 10]
best = actions[value.index(max(value))]
print("Best Action:", best)

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
```

4. Cryptocurrency Trading System

Actions

- Buy
- Sell
- Hold

Reward Function Challenges

- Delayed profit
- Highly volatile market
- Transaction costs
- Risk management

Poor Reward Design Effects

- Excessive trading
- Financial losses
- Risky decisions
- Unstable learning

A properly designed reward function helps maximize long-term profit instead of short-term gains.

The image shows a Python script in a text editor and its execution in the IDLE Shell. The script defines a list of prices, a list of actions, and a reward function. The shell shows the execution of the script, including the reward function and the actions taken.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
File Edit Format Run Options Window Help
prices = [100, 105, 98]
actions = ["Buy", "Sell", "Hold"]

reward = prices[-1] - prices[0]
print("Reward:", reward)
print("Action:", actions[2])

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
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>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
```

5. Smart Grid Electricity Distribution

RL Framework

State

- Electricity demand
- Available power
- Battery level
- Weather conditions

Action

- Increase supply
- Decrease supply
- Store energy
- Use renewable energy

Reward

- +10 for balanced power distribution
- +15 for energy savings
- -10 for power outage
- -15 for energy wastage

The RL agent learns efficient electricity distribution while minimizing wastage and maintaining stable power supply.

The image shows a Python script in a text editor and its execution in the IDLE Shell. The script defines a state variable and a function to determine the action based on the state. The shell shows the execution of the script, including the state variable and the actions taken.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
File Edit Format Run Options Window Help
state = "High Demand"

if state == "High Demand":
    action = "Increase Supply"
else:
    action = "Store Energy"

print(action)

IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
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>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
```

6. Healthcare Medication Adjustment

RL Framework

State

- Patient health condition
- Blood pressure
- Heart rate
- Previous medication response

Action

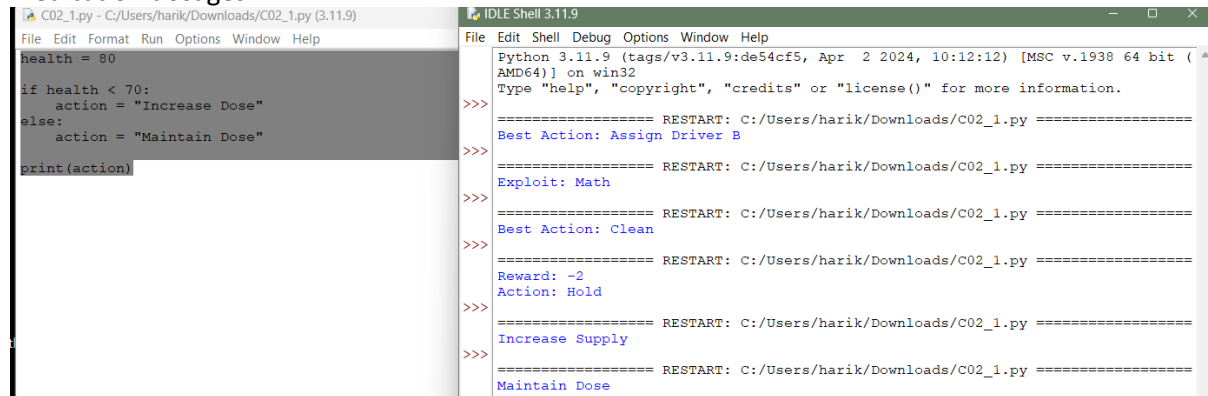
- Increase dosage
- Decrease dosage
- Maintain dosage
- Change medication

Reward

- +20 for improved patient health
- +10 for stable condition
- -20 for adverse effects
- -15 for incorrect dosage

Learning

The RL agent continuously learns from patient responses and recommends safer, personalized medication dosages.



The image shows a Python script in a text editor and its execution in an IDLE Shell. The script defines a health variable and a conditional action based on health status. The shell output shows the script being executed, with restarts and actions like 'Assign Driver B', 'Math', 'Clean', 'Hold', 'Supply', and 'Dose'.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
health = 80
if health < 70:
    action = "Increase Dose"
else:
    action = "Maintain Dose"
print(action)
```

```
IDLE Shell 3.11.9
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
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>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Maintain Dose
>>>
```

7. Drone Delivery System

Exploration

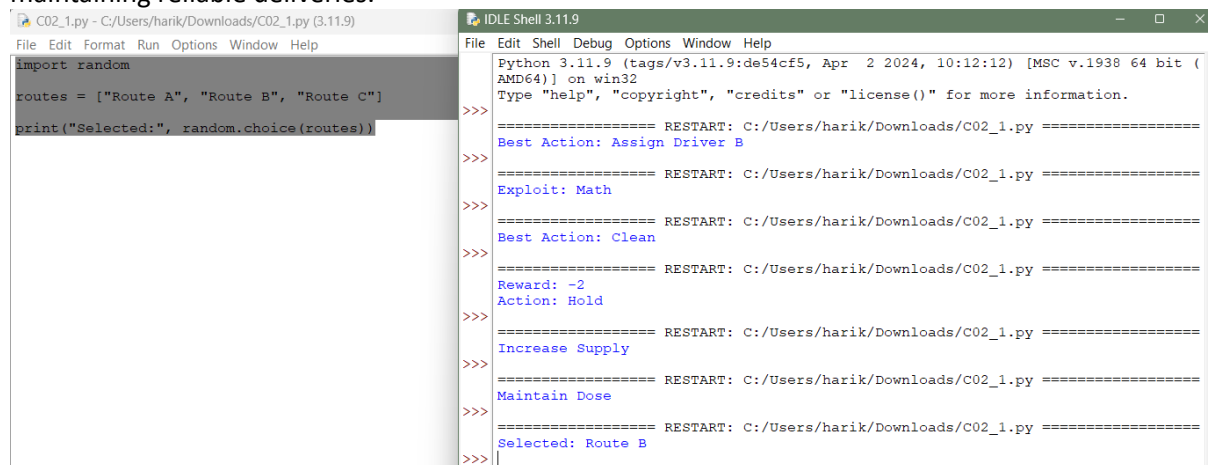
- Tests new delivery routes.
- Finds shorter paths.
- Adapts to changing environments.

Exploitation

- Uses previously successful routes.
- Reduces delivery time.
- Saves battery power.

Effect on Route Optimization

A balance between exploration and exploitation enables drones to discover efficient routes while maintaining reliable deliveries.



The image shows a Python script in a text editor and its execution in an IDLE Shell. The script imports random and defines a list of routes. The shell output shows the script being executed, with restarts and actions like 'Assign Driver B', 'Math', 'Clean', 'Hold', 'Supply', 'Dose', and 'Selected: Route B'.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
import random
routes = ["Route A", "Route B", "Route C"]
print("Selected:", random.choice(routes))
```

```
IDLE Shell 3.11.9
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
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>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Maintain Dose
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Selected: Route B
>>>
```

8. Manufacturing Robot

RL Framework

State

- Robot position
- Part availability
- Machine status
- Assembly progress

Action

- Pick component
- Assemble part
- Move robot arm
- Inspect product

Policy

The policy selects the best assembly action based on the current production state.

Value Function

The value function predicts future rewards, enabling the robot to reduce assembly time, improve product quality, and increase production efficiency.

```

C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9)
File Edit Format Run Options Window Help
tasks = ["Pick", "Assemble", "Inspect"]
values = [4, 9, 6]
print("Best Task:", tasks[values.index(max(values))]) >>>

IDLE Shell 3.11.9
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Maintain Dose
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Selected: Route B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Task: Assemble
>>>

```

9. Online Advertising System

Reward Design Challenges

- Delayed customer purchases
- Click fraud
- Balancing clicks and conversions
- Changing user interests

Impact of Poor Reward Design

- Low-quality advertisements
- Reduced click-through rate
- Wasted advertising budget
- Poor customer engagement

A well-designed reward function maximizes meaningful user interactions rather than only increasing clicks.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9) IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
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Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Maintain Dose
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Selected: Route B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Task: Assemble
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: 10
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: 10
>>>
|
```

10. Smart Irrigation System

RL Framework

State

- Soil moisture
- Weather forecast
- Water availability
- Crop growth stage

Action

- Turn irrigation ON
- Turn irrigation OFF
- Increase water flow
- Reduce water flow

Reward

- +15 for healthy crop growth
- +10 for water conservation
- -15 for overwatering
- -10 for underwatering

The RL agent learns the optimal irrigation schedule to conserve water while maintaining healthy crop growth.

```
C02_1.py - C:/Users/harik/Downloads/C02_1.py (3.11.9) IDLE Shell 3.11.9
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Assign Driver B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Exploit: Math
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Action: Clean
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: -2
Action: Hold
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Increase Supply
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Maintain Dose
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Selected: Route B
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Best Task: Assemble
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: 10
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Reward: 10
>>>
===== RESTART: C:/Users/harik/Downloads/C02_1.py =====
Water ON
>>>
```